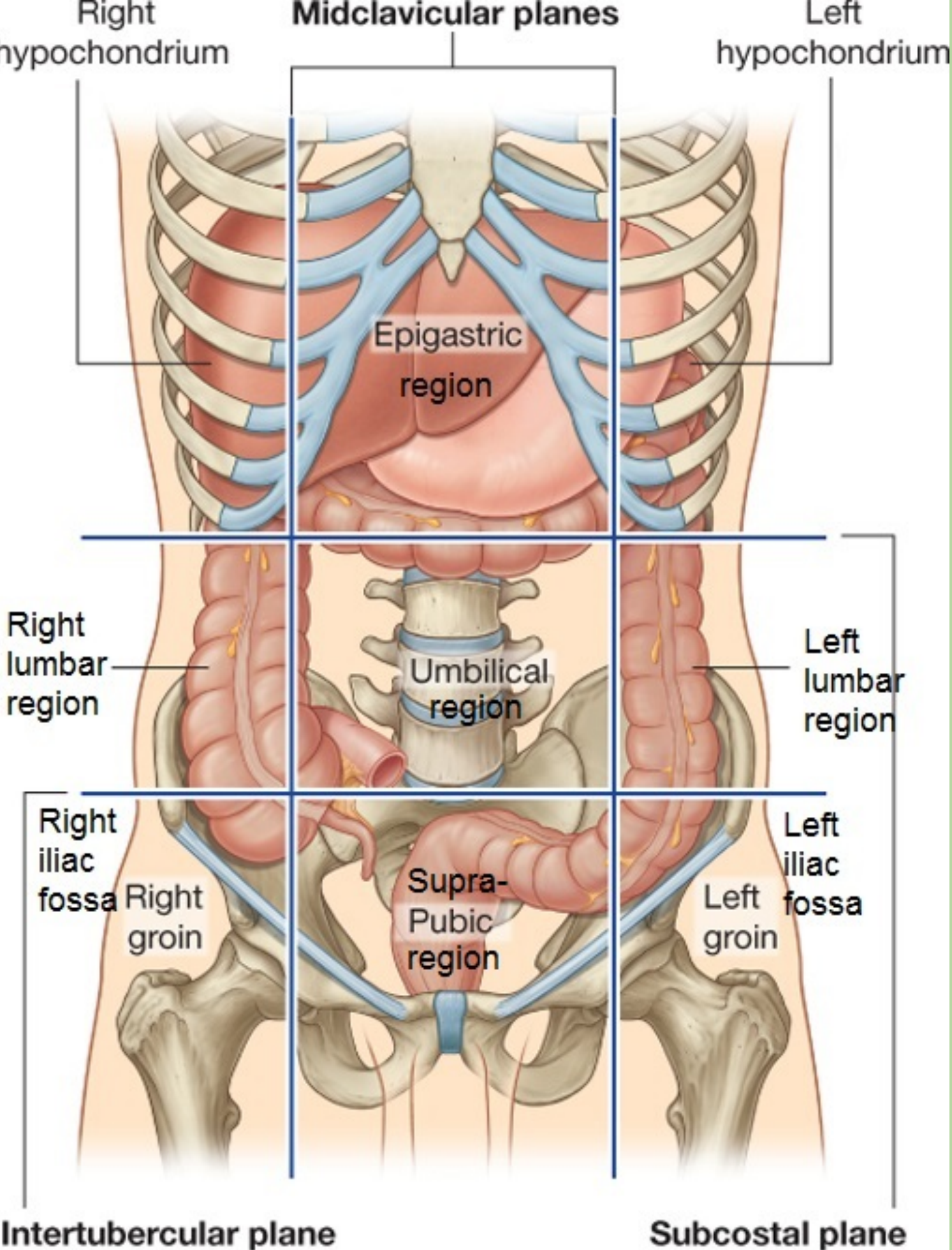
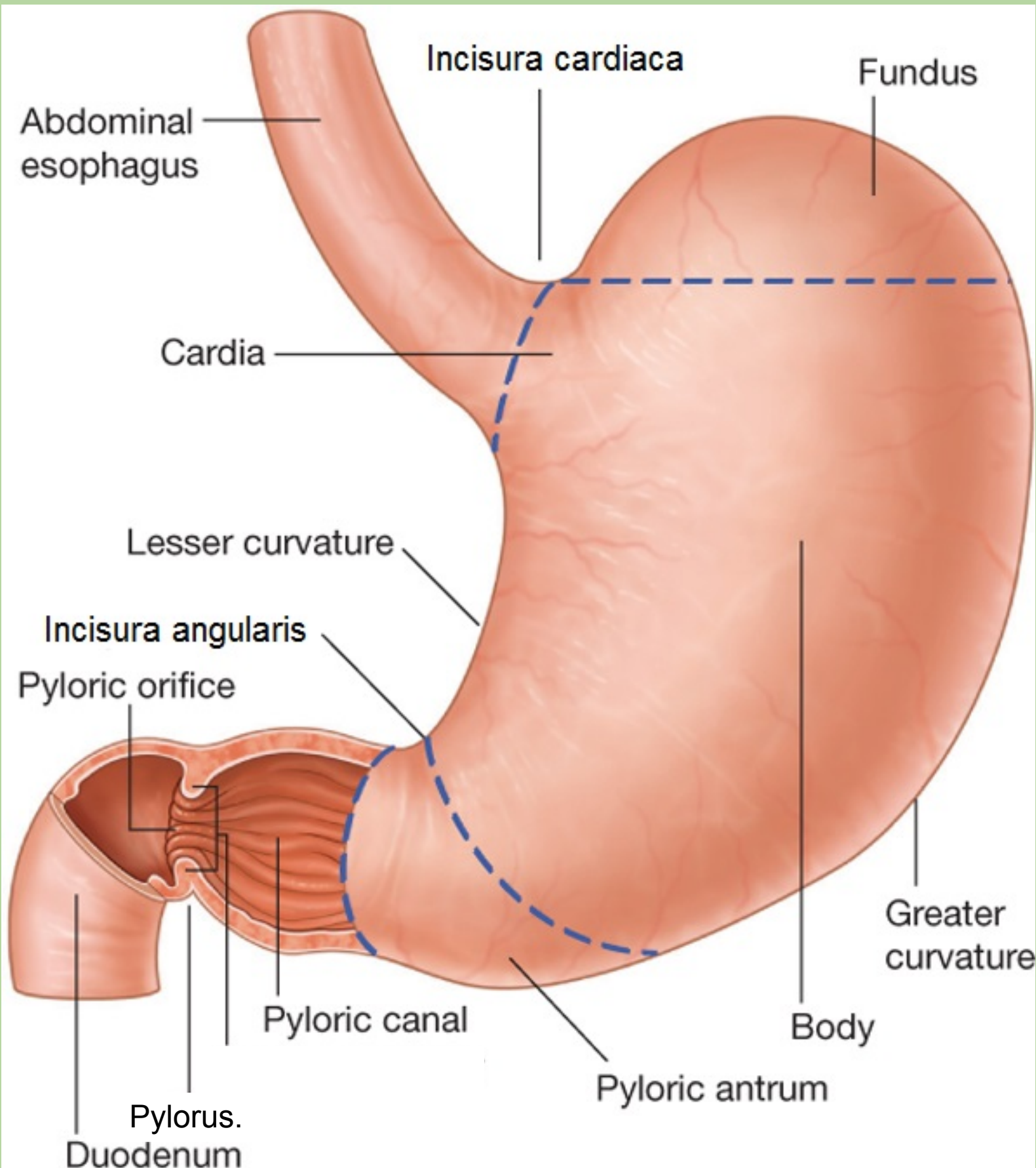




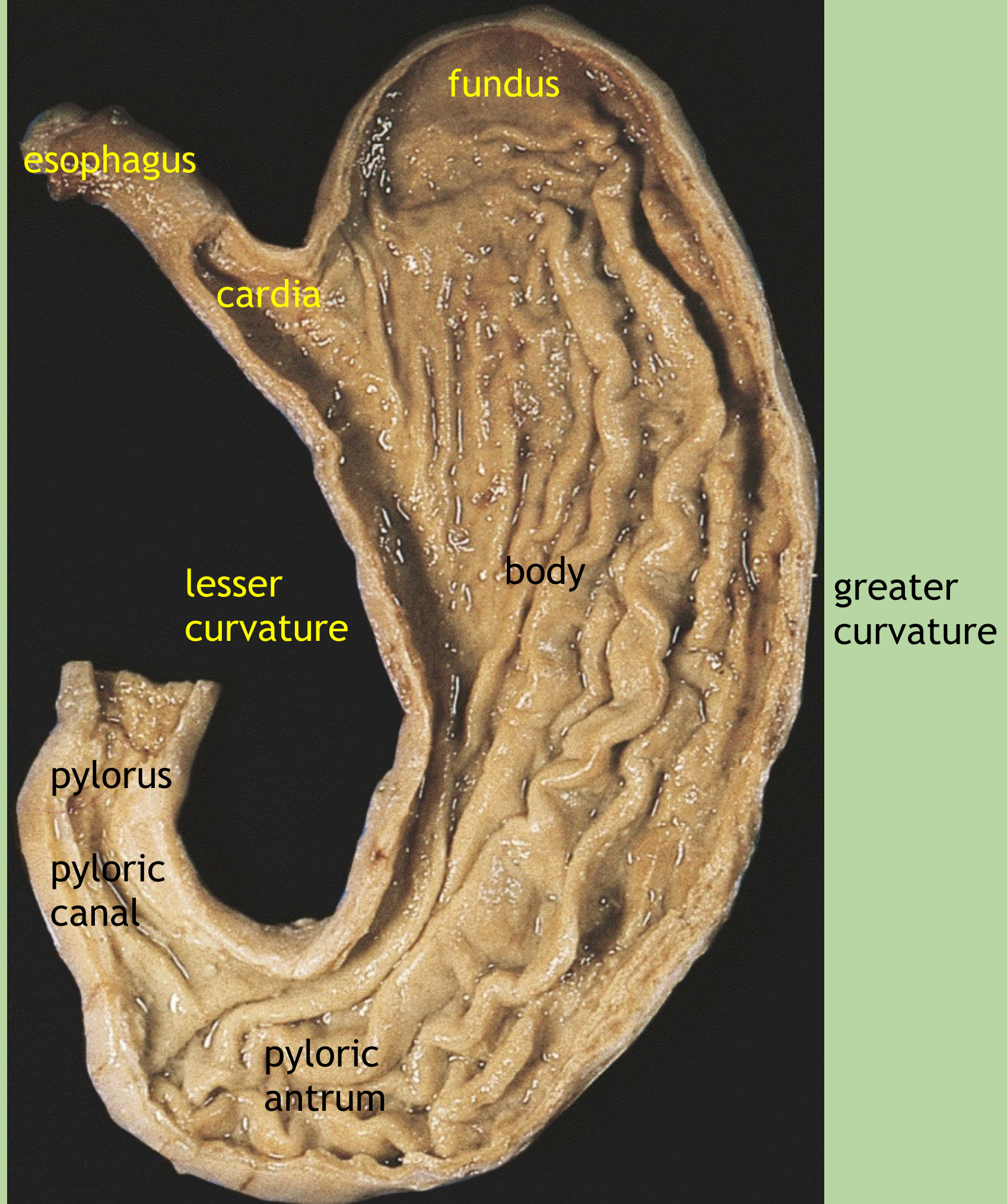
Location of the stomach

- Left hypochondrium
- Epigastrium
- (Umbilical region)

Parts of the stomach



- Cardia
- Fundus
- Body
- Pyloric antrum
- Pylorus





Interior of the stomach

Prominent gastric mucosal folds (rugae) increase the surface area. They are directed longitudinally toward the pylorus.

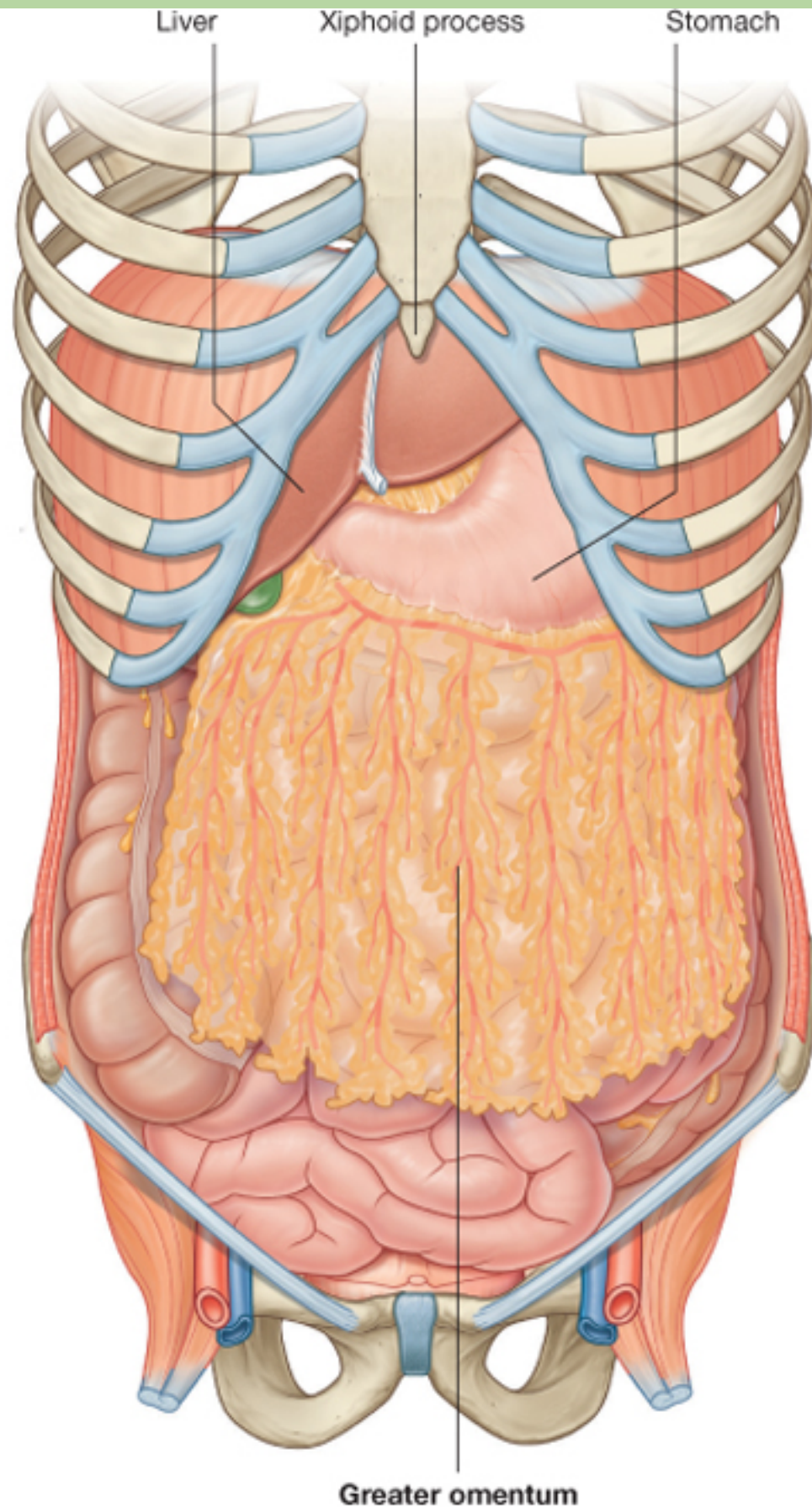
Average capacity of stomach:

in newborns: 15 ml

during puberty: 1000 ml

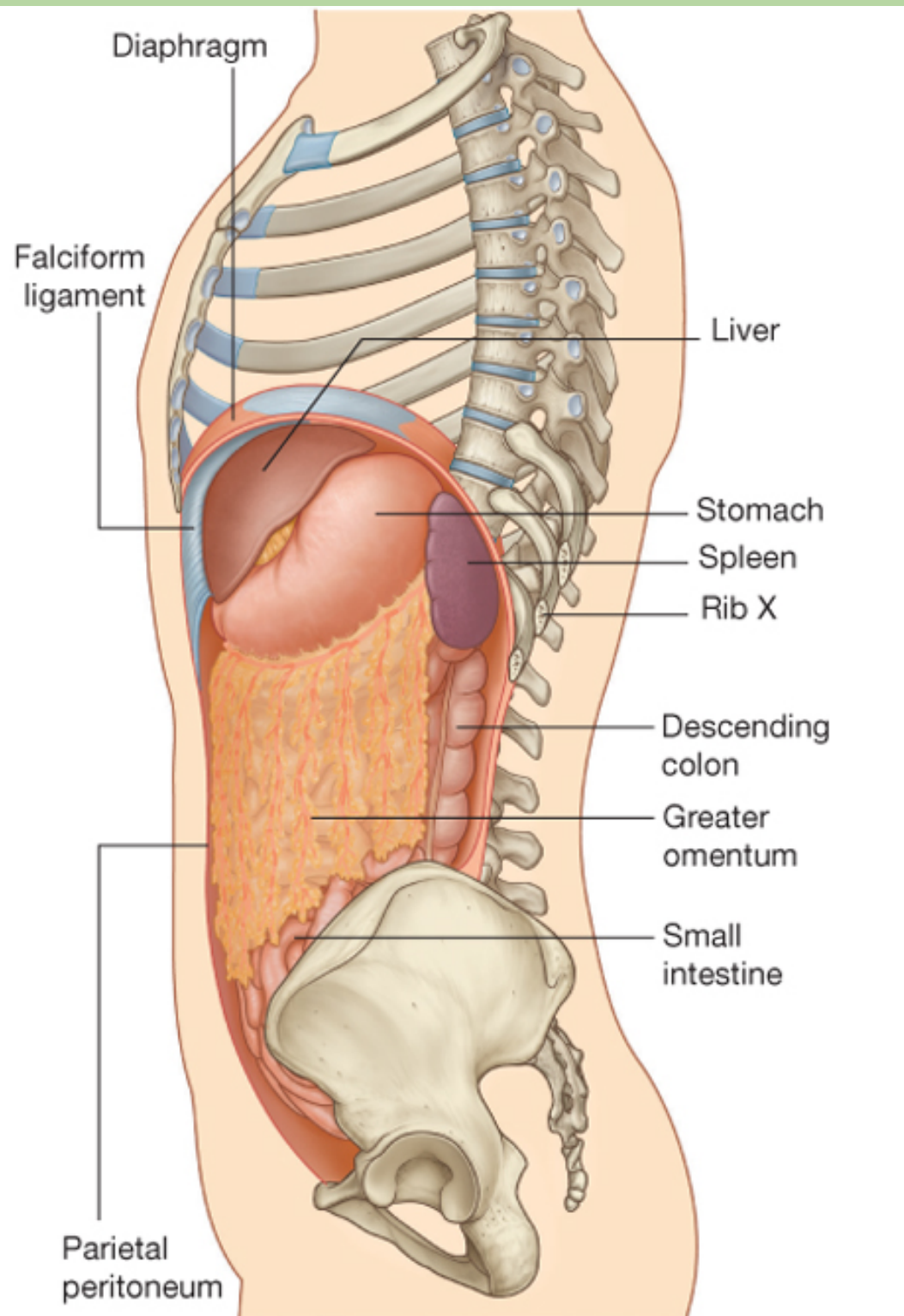
in adults: 1500ml

Contact areas of adjacent organs – anterior stomach wall

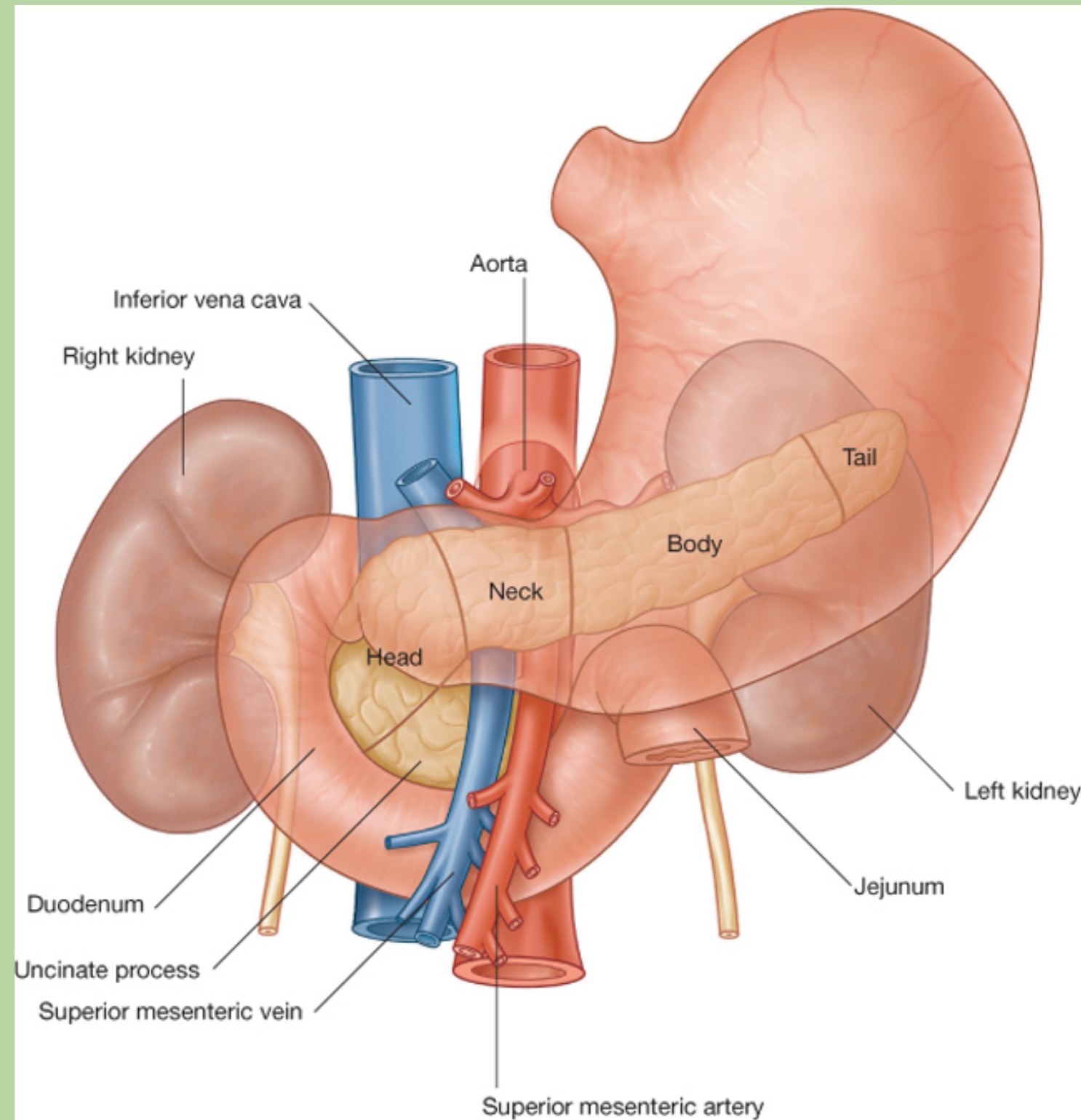


- Phrenic surface
- Hepatic surface
- Epigastric surface (facies libera)

Contact areas of adjacent organs – posterior stomach wall



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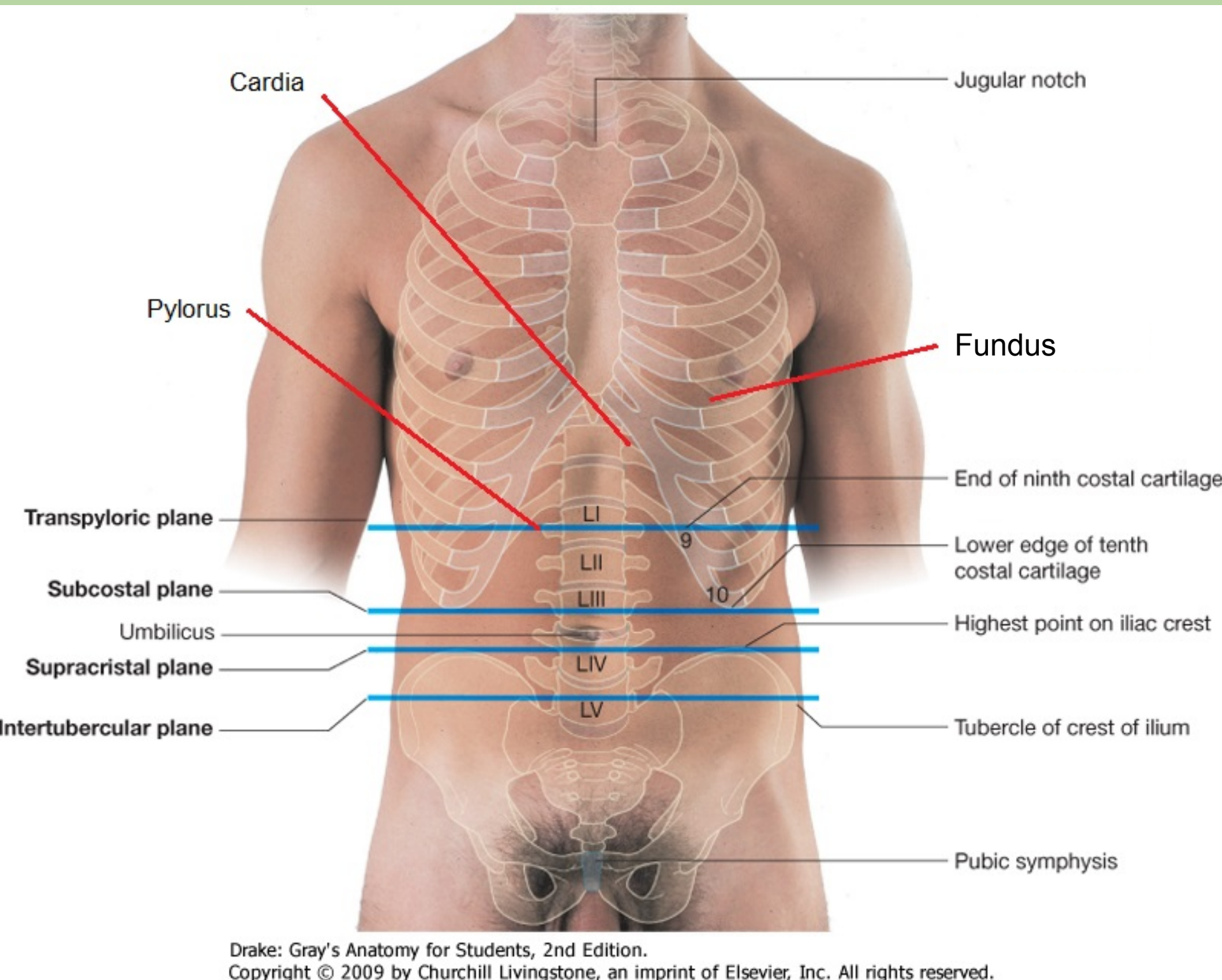


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- Splenic, Colomesocolic, Pancreatic, Renal, Suprarenal surfaces



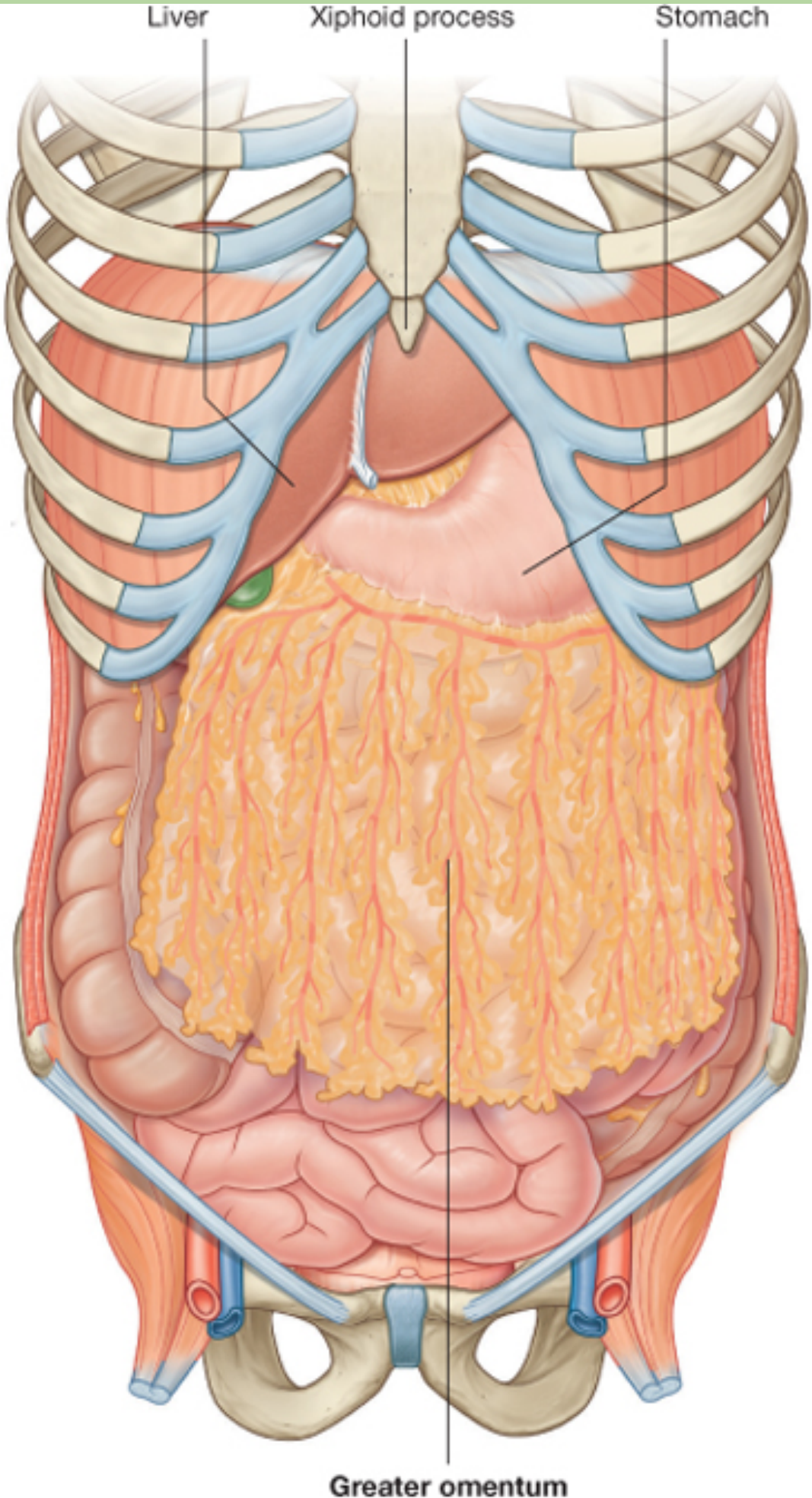
Topography



Cardia: 2 cm to the left from the midline behind the 7th costal cartilage at the level of left edge of T11 vertebra. (relatively constant)

Tip of the fundus: at the level of 5th intercostal space near to the mid-inguinal line. (relatively constant)

Pylorus: 2-3 cm to the right from the midline at the level of L1 vertebra. (depends on food volume)



Peritoneal relationships

Stomach is an **intraperitoneal** organ.

Greater omentum: from the greater curvature.

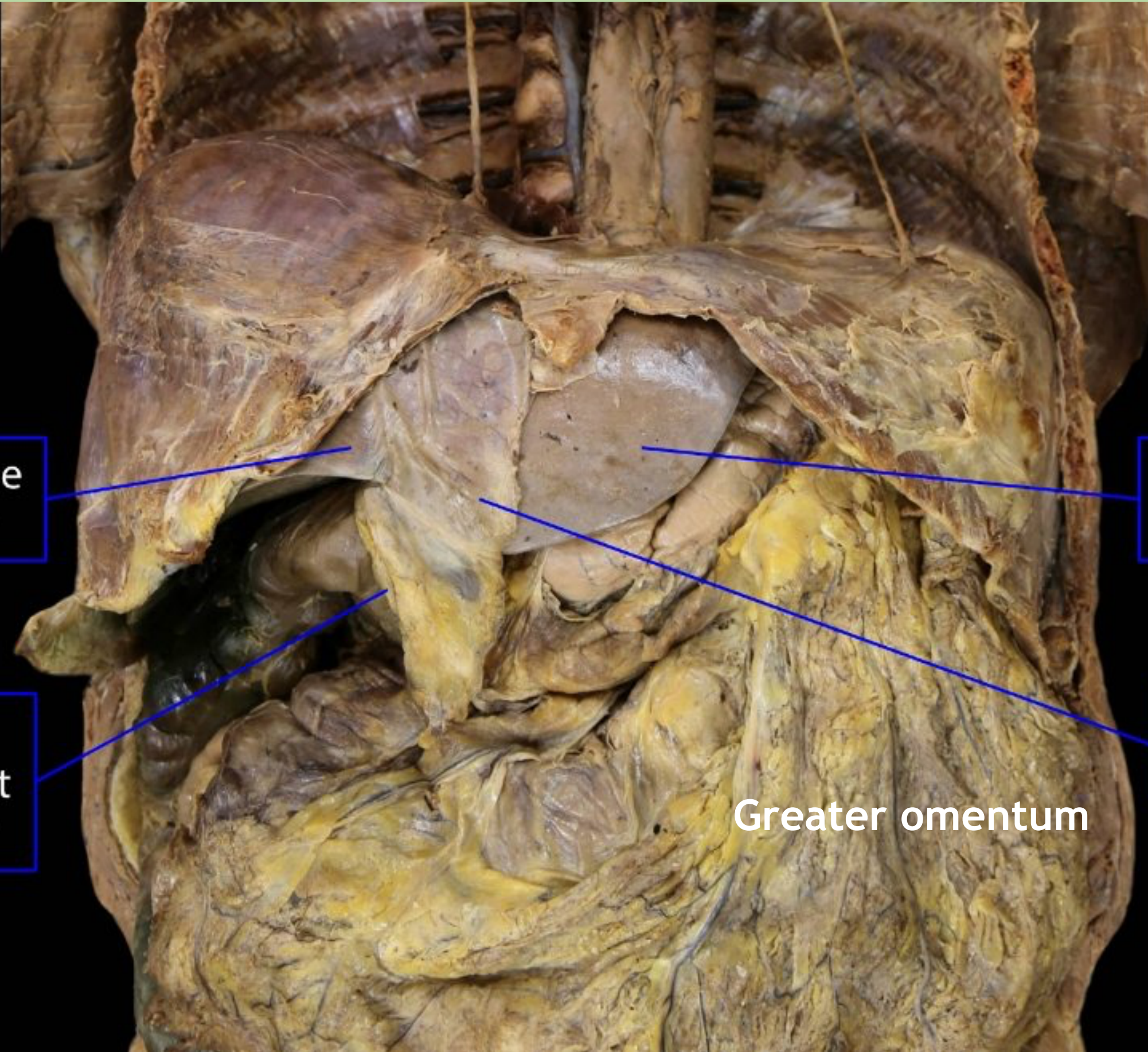
Right lobe
of liver

Left lobe
of liver

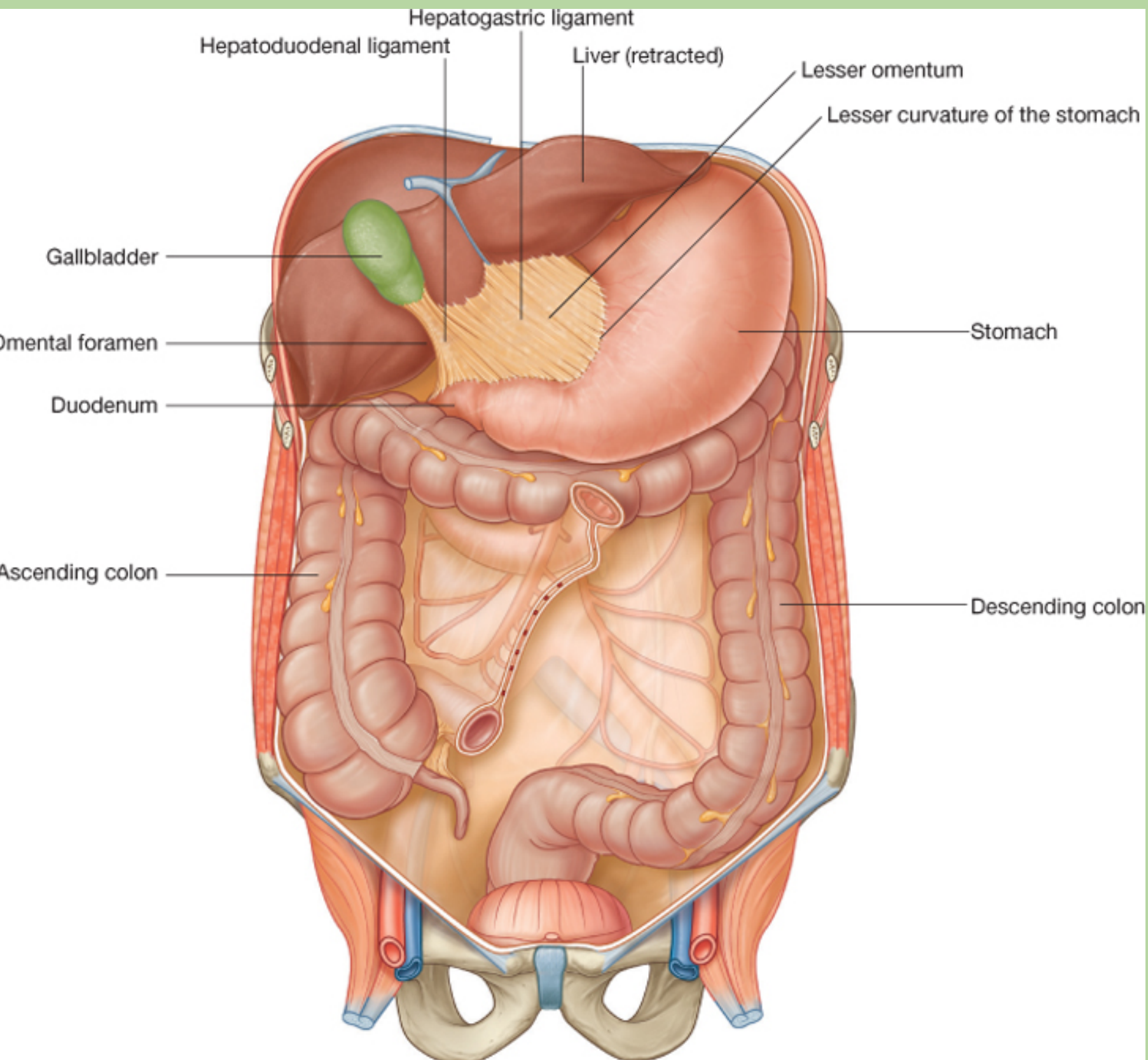
Round
ligament
of liver

Falciform
ligament

Greater omentum



Peritoneal relationships

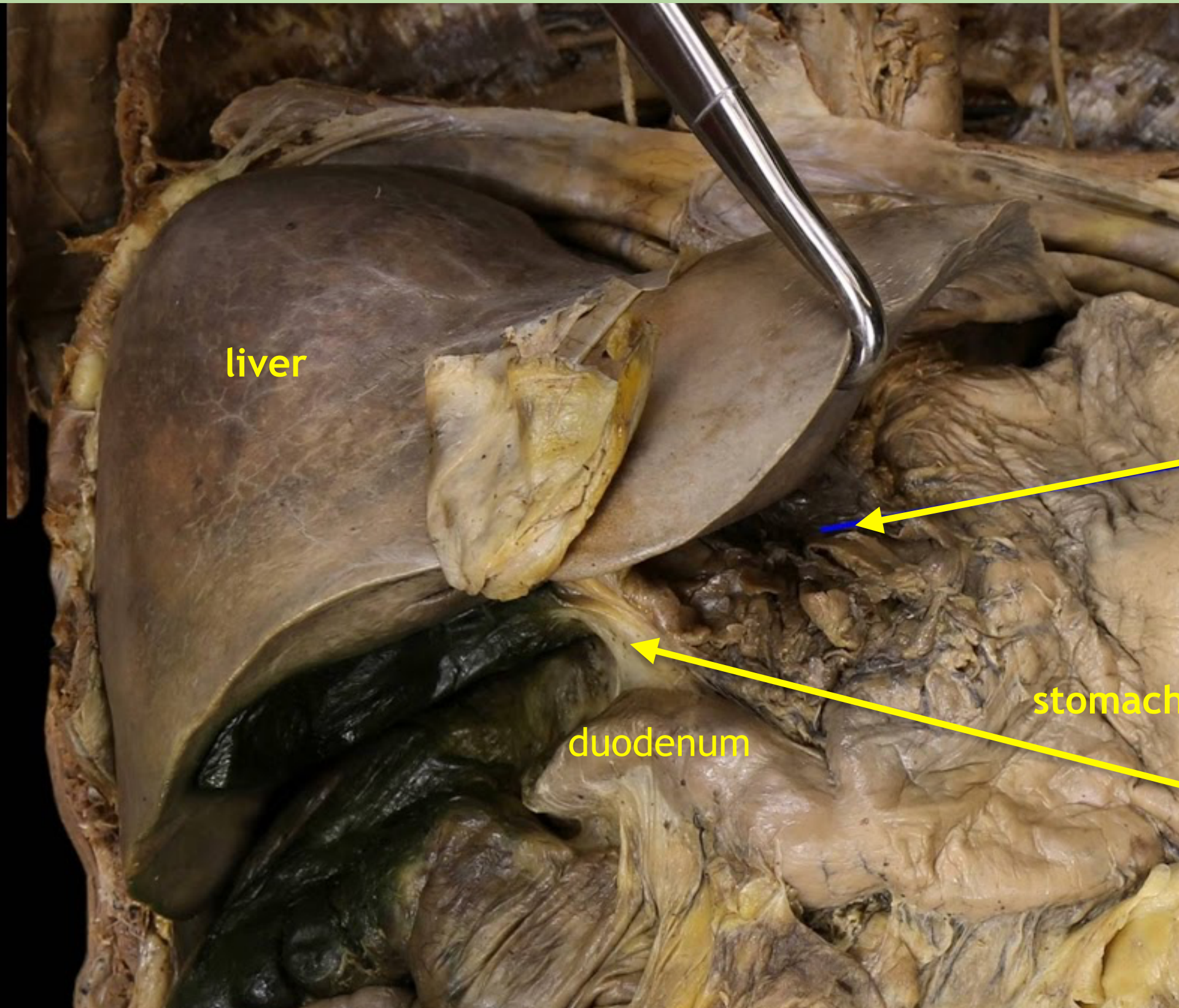


Hepatogastric ligament:
between the lesser
curvature and liver

Lesser omentum:
hepatogastric ligament +
hepatoduodenal ligament;

posterior to it: omental
bursa

Peritoneal relationships



***Lesser omentum:
hepatogastric ligament
+ hepatoduodenal
ligament.***

hepatogastric
ligament

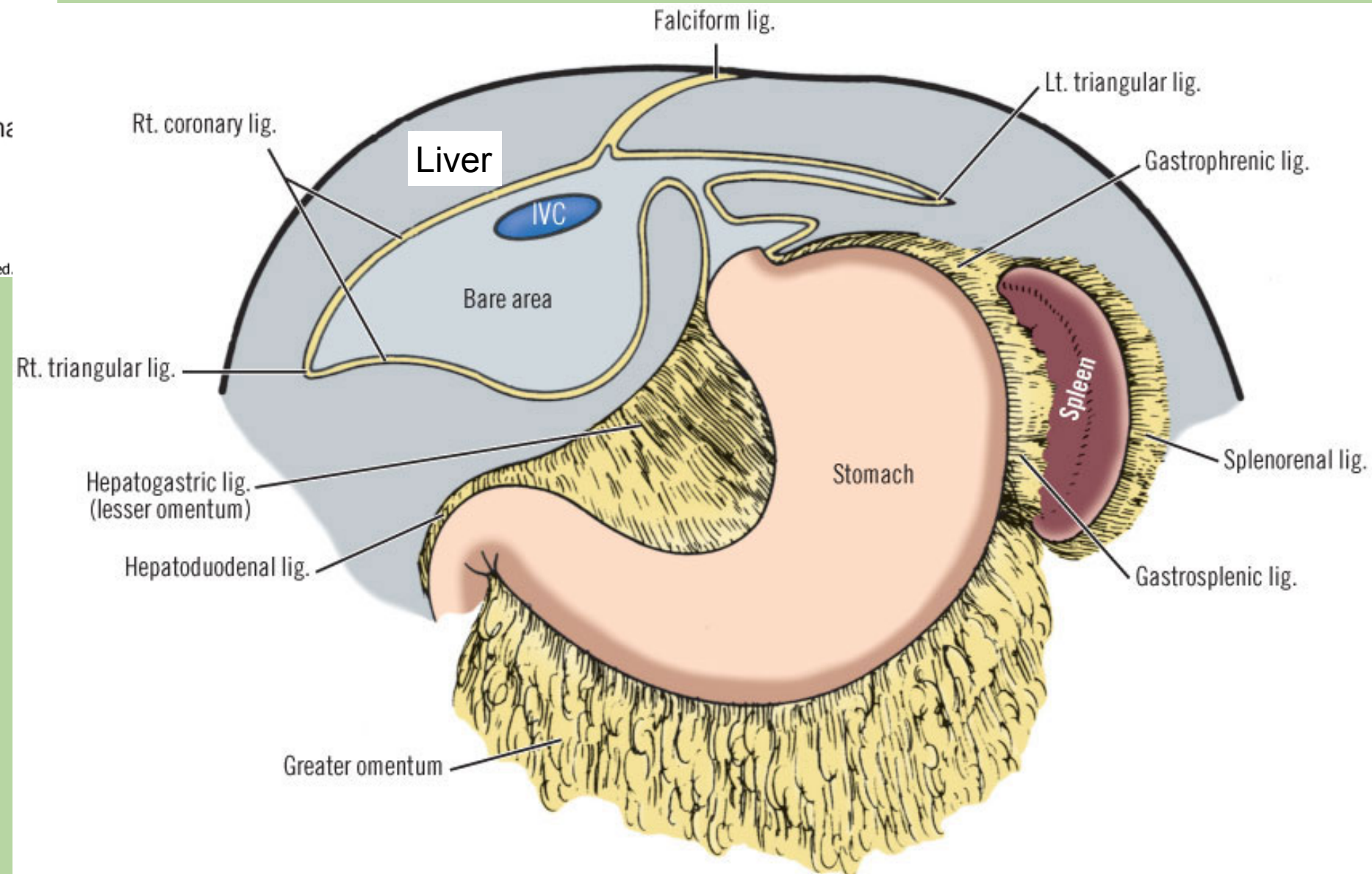
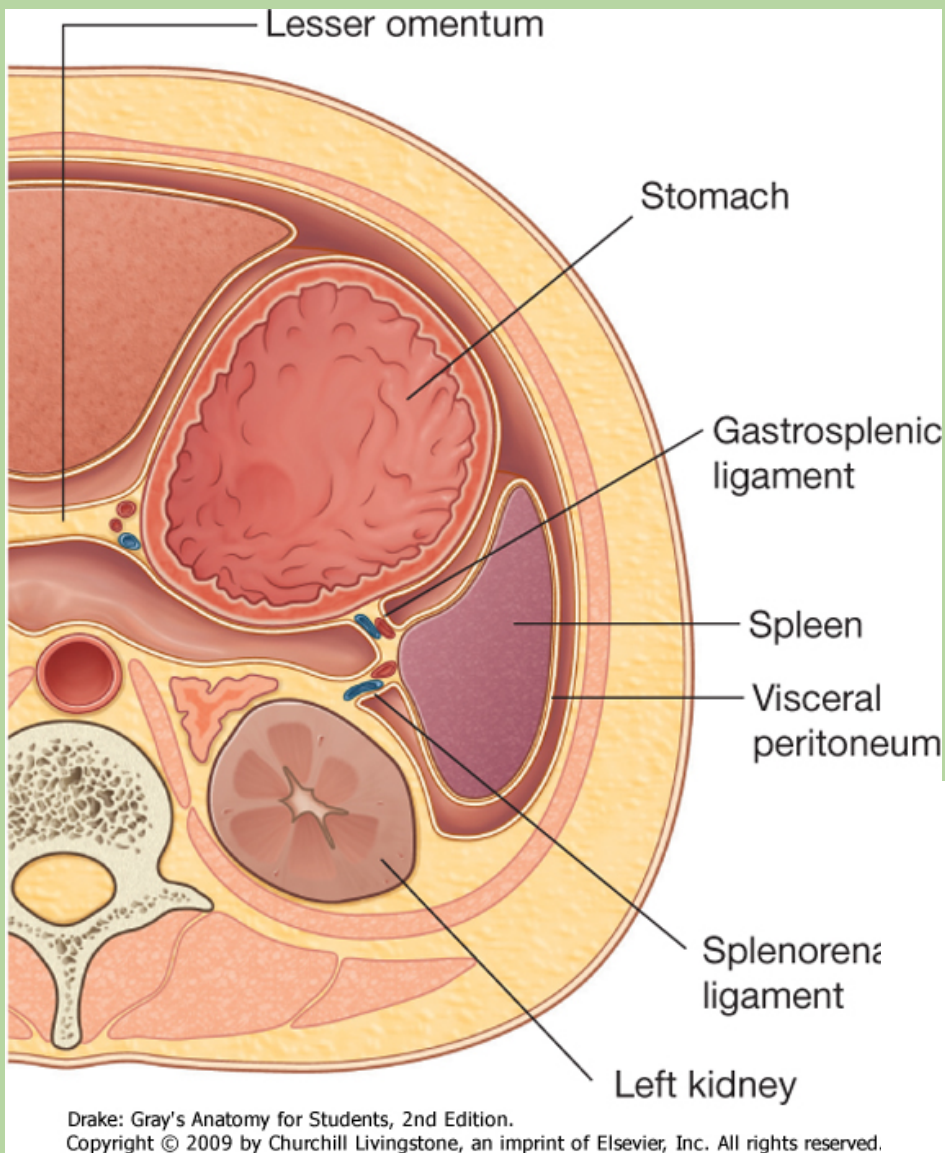
hepatoduodenal
ligament
*(between the first part
of duodenum and liver)*

Peritoneal relationships

Gastrophrenic ligament: from the upper portion of greater curvature to the diaphragm

Gastrosplenic ligament: from the lateral portion of greater curvature to the spleen

Gastrocolic ligament: between the greater curvature and transverse colon





Liver

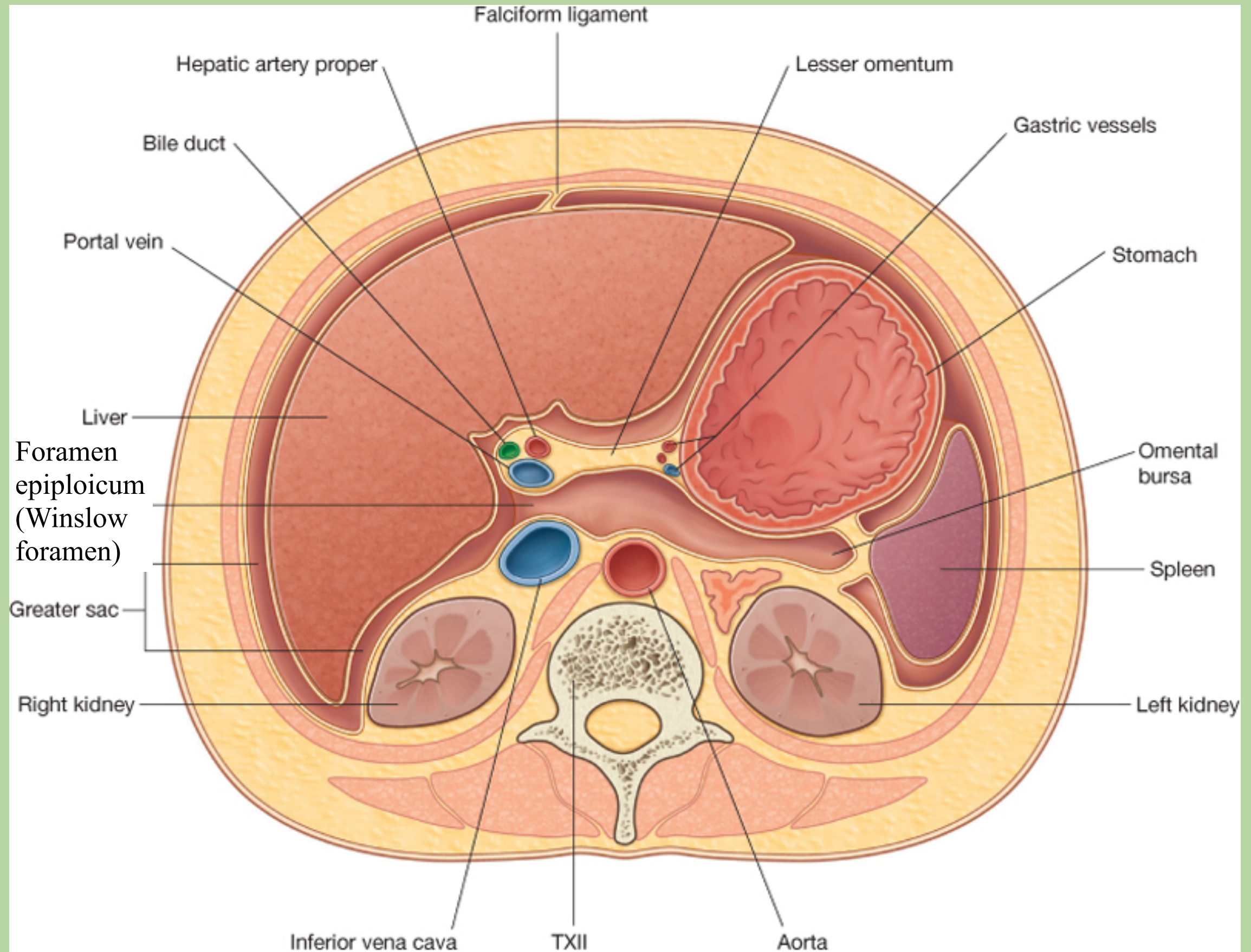
Diaphragm

Stomach

SPLLEN

Gastrosplenic ligament
with short gastric vessels

The largest peritoneal recess: omental bursa



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Vasculature - arteries

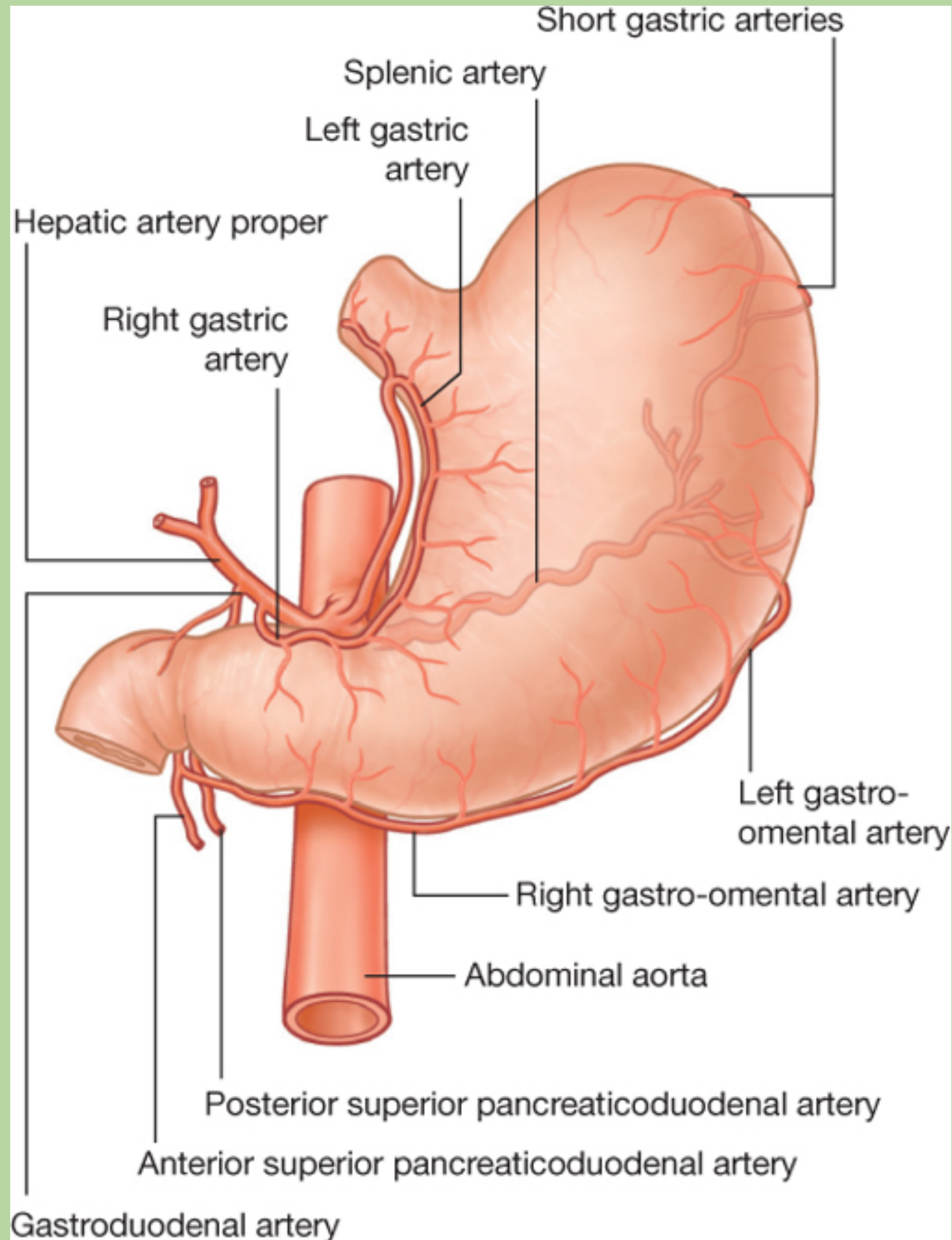
ON THE LESSER CURVATURE (in the hepatogastric ligament):

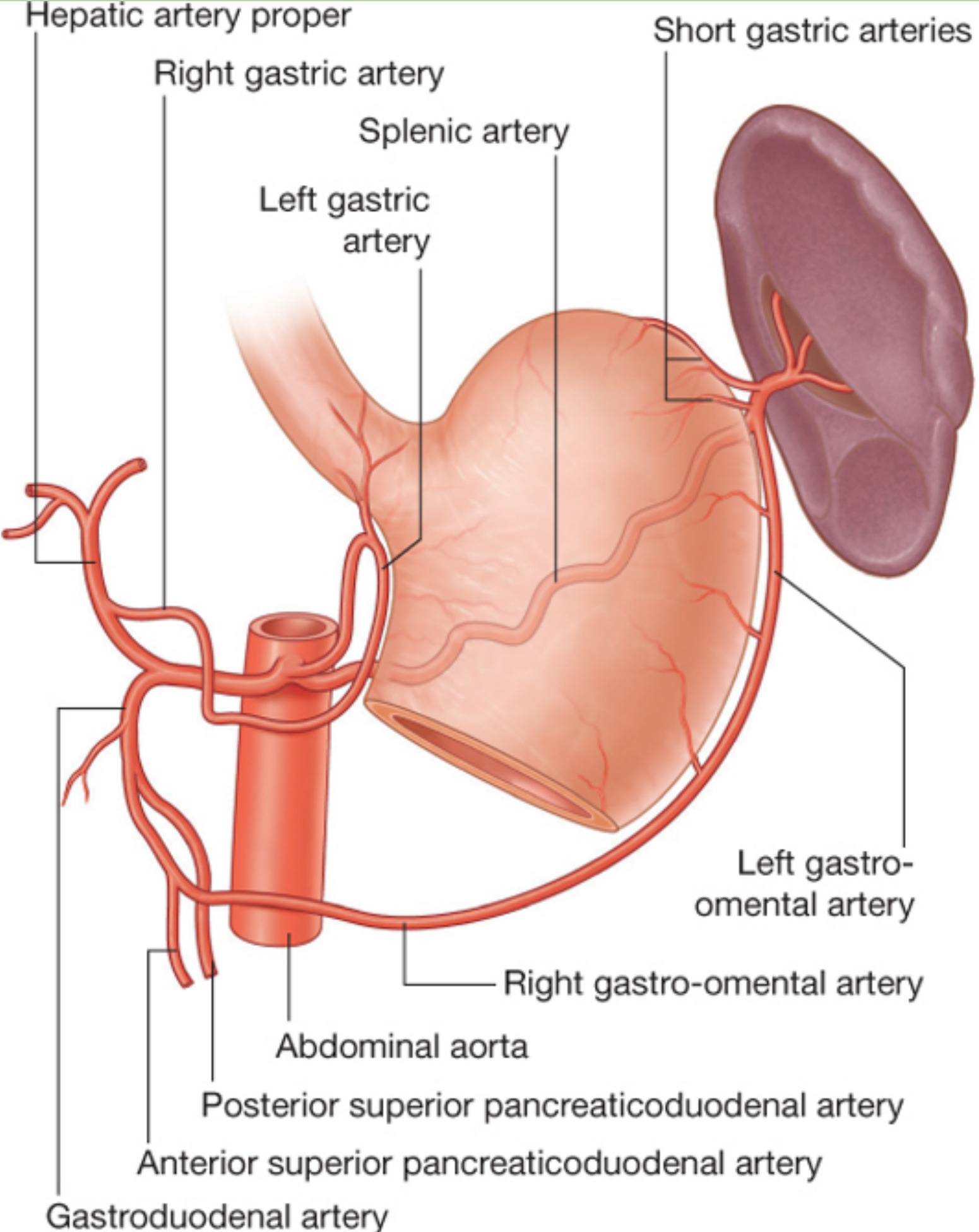
Left gastric artery: a direct branch of the celiac trunk.

Right gastric artery: arises from hepatic artery proper. (Its origin often varies: the most common alternative is the common hepatic.)

FUNDUS:

Short gastric arteries: they arise from the splenic artery and situate in the gastrosplenic ligament.



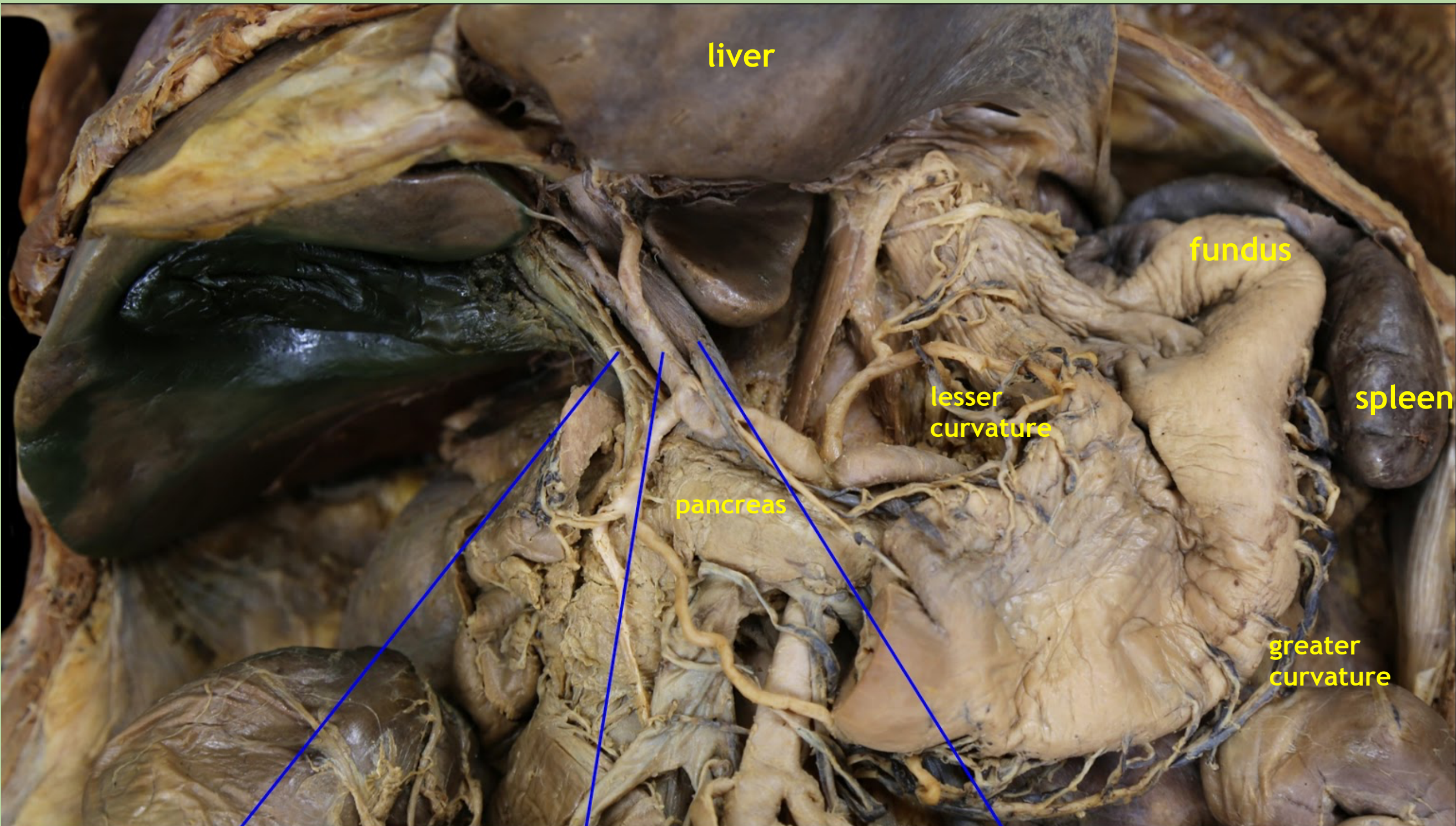


Vasculature - arteries

ON THE GREATER CURVATURE (in the greater omentum):

Left gastroepiploic (gastro-omental) artery: branch of the splenic artery.

Right gastroepiploic (gastro-omental) artery: originates from the gastroduodenal artery.



liver

fundus

spleen

lesser
curvature

pancreas

greater
curvature

Bile duct

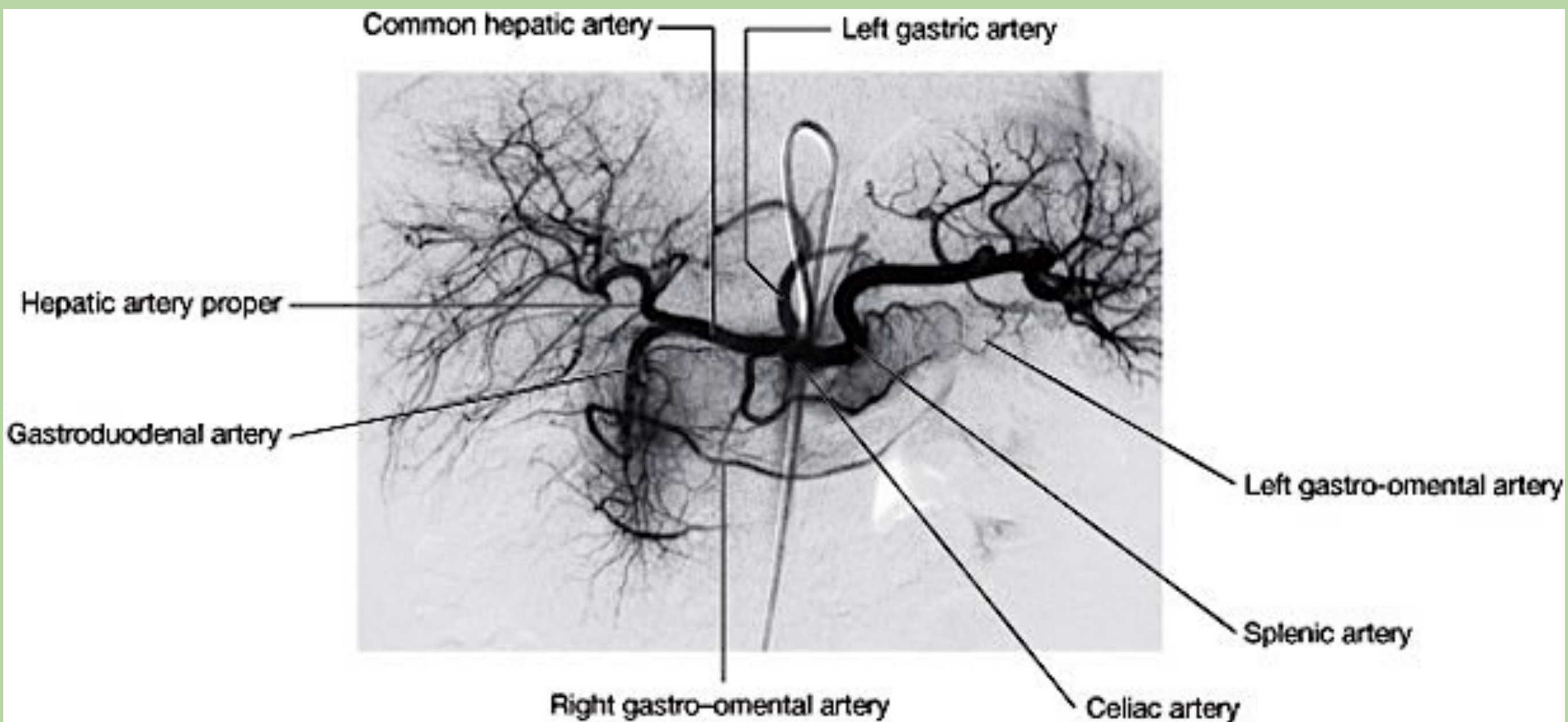
Hepatic artery
proper

Hepatic
portal vein

Hepatoduodenal lig. contents, anterior

BlueLink

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Vasculature - veins

ON THE LESSER CURVATURE:

Left gastric and right gastric veins: drain into the hepatic portal vein.

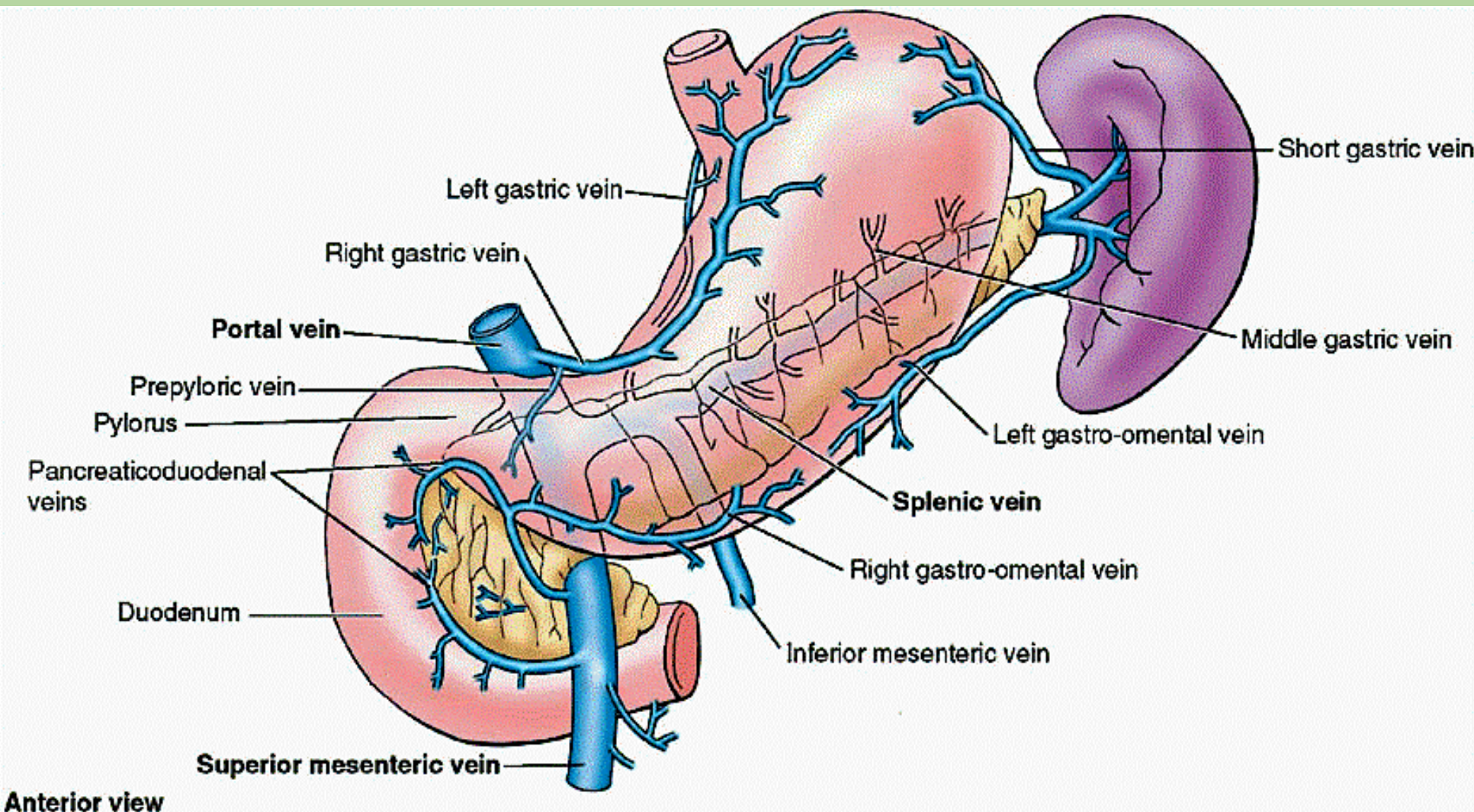
ON THE GREATER CURVATURE:

Left gastroepiploic (gastro-omental) vein: drains into the splenic vein.

Right gastroepiploic vein: drains into the superior mesenteric vein.

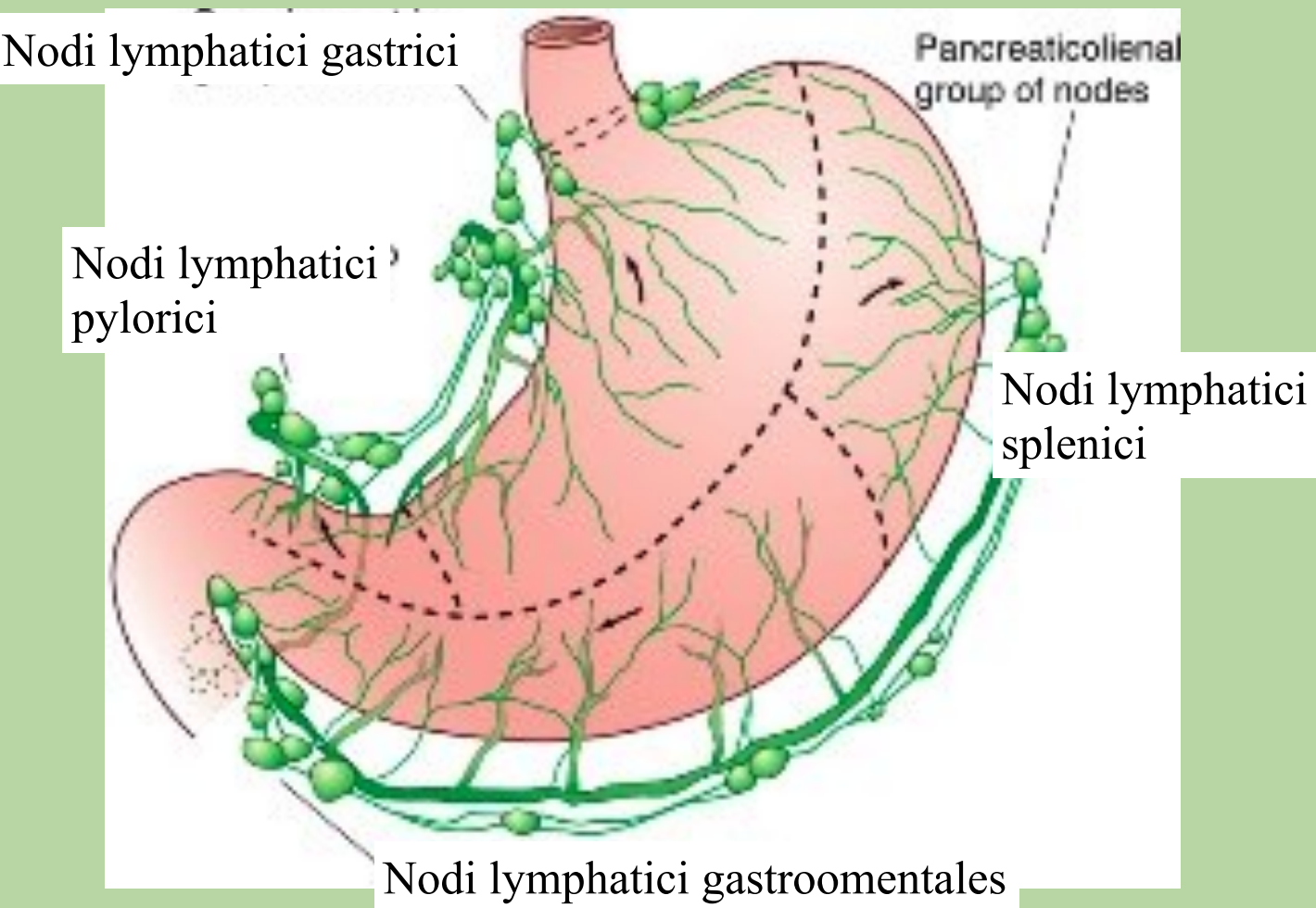
FUNDUS:

Short gastric veins: drains into the splenic vein.



Prepyloric vein: ascends over the pylorus to the right gastric vein. (Surgeons use it for identifying the pylorus.)

Lymphatic drainage of the stomach



Gastric lymphatic vessels accompany the arteries along the greater and lesser curvatures. They drain lymph from its anterior and posterior surfaces toward its curvatures, where the gastric and gastro-omental lymph nodes are located.

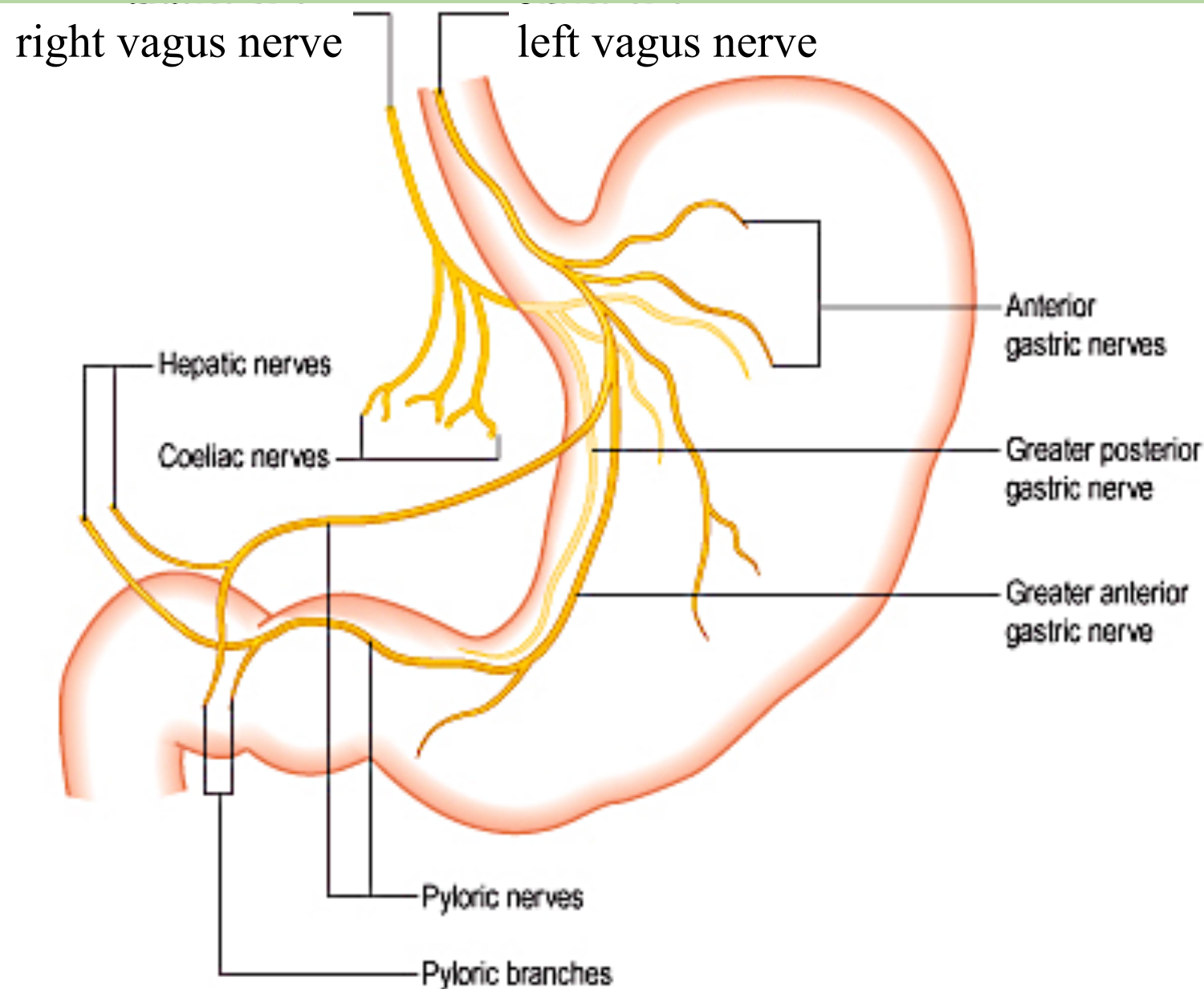
Lymph from superior two thirds of stomach drained toward the ***gastric and pancreaticosplenic lymph nodes***.

Lymph from the right two thirds of the inferior third drained toward ***pyloric lymph nodes***.

Lymph from the left one third of greater curvature drained toward the ***gastro-omental and pancreaticoduodenal lymph nodes***.

Innervation of the stomach

Parasympathetic innervation:
left vagus nerve – anterior side
right vagus nerve – posterior side



The parasympathetic gastric supply is secretomotor to the gastric mucosa and motor to the gastric musculature. It is responsible for coordinated relaxation of the pyloric sphincter during gastric emptying.

Sympathetic innervation: from T6-9 ganglia via the *greater splanchnic nerve via coeliac plexus*.

The gastric sympathetic nerves are vasoconstrictor to the gastric vasculature and inhibitory to gastric musculature. The sympathetic supply to the pylorus is motor, and brings about pyloric constriction.

Main functions of the stomach

- Pepsinogen production
- Mucous secretion – gastric mucosa barrier
- HCL and intrinsic factor production
- Gastrin, somatostatin secretion
- Motor function: peristalsis
- Absorption: some drugs (aspirin), ethanol, caffeine

A

Superior part of duodenum Esophagus

Pyloric antrum

Fundus of stomach



Descending part of duodenum

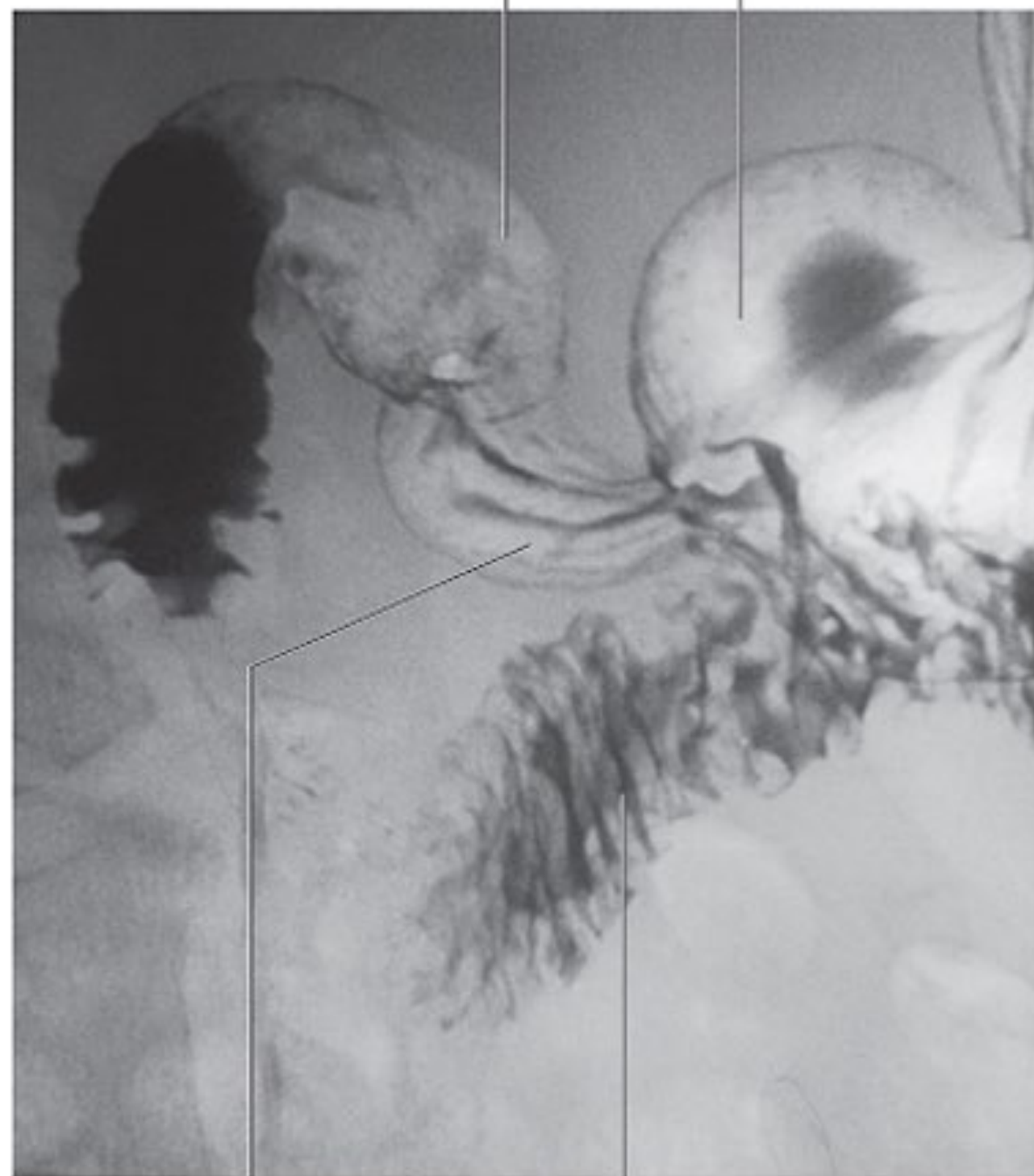
Body of stomach

Duodenal jejunal flexure

B

Normal duodenal cap

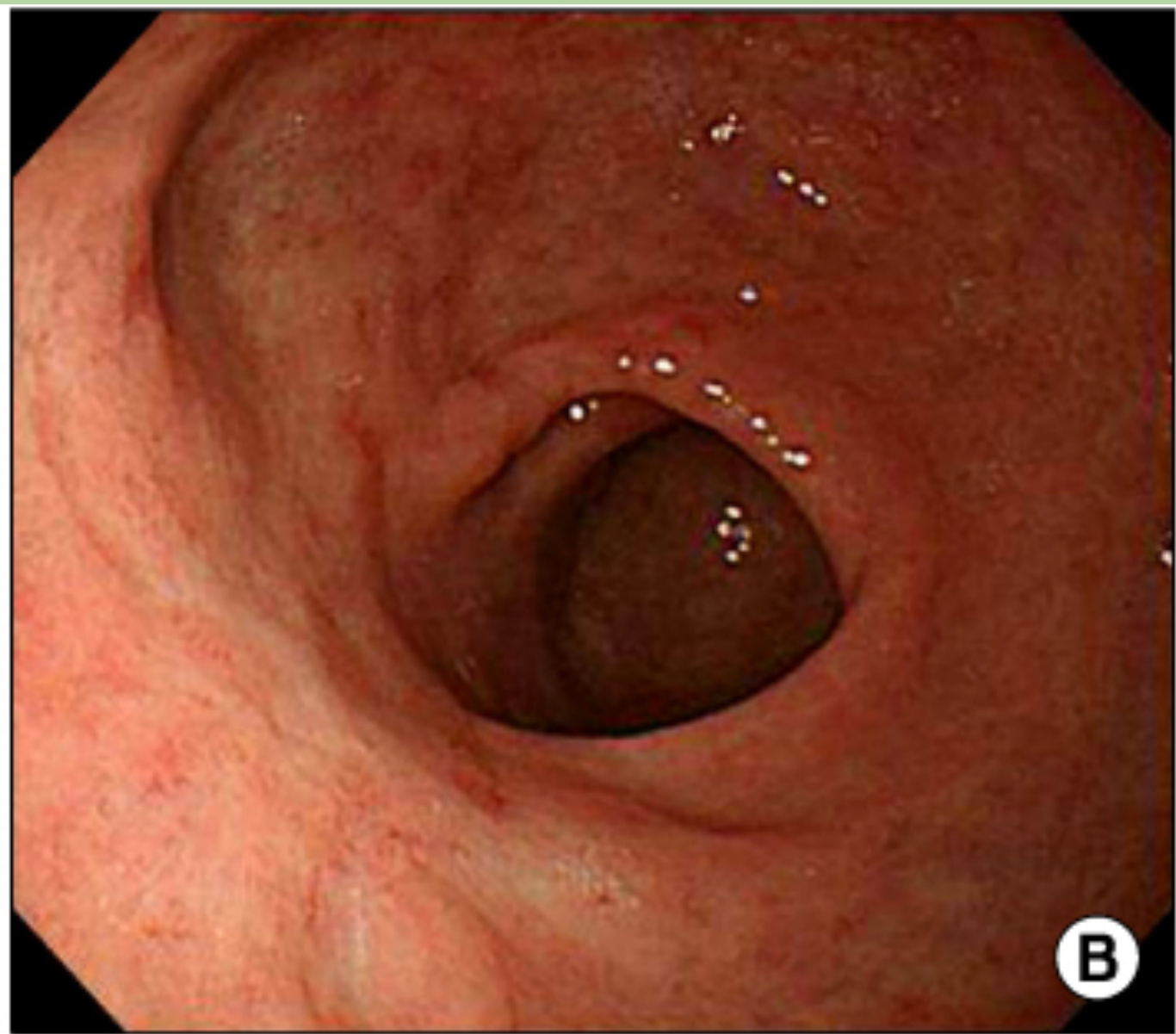
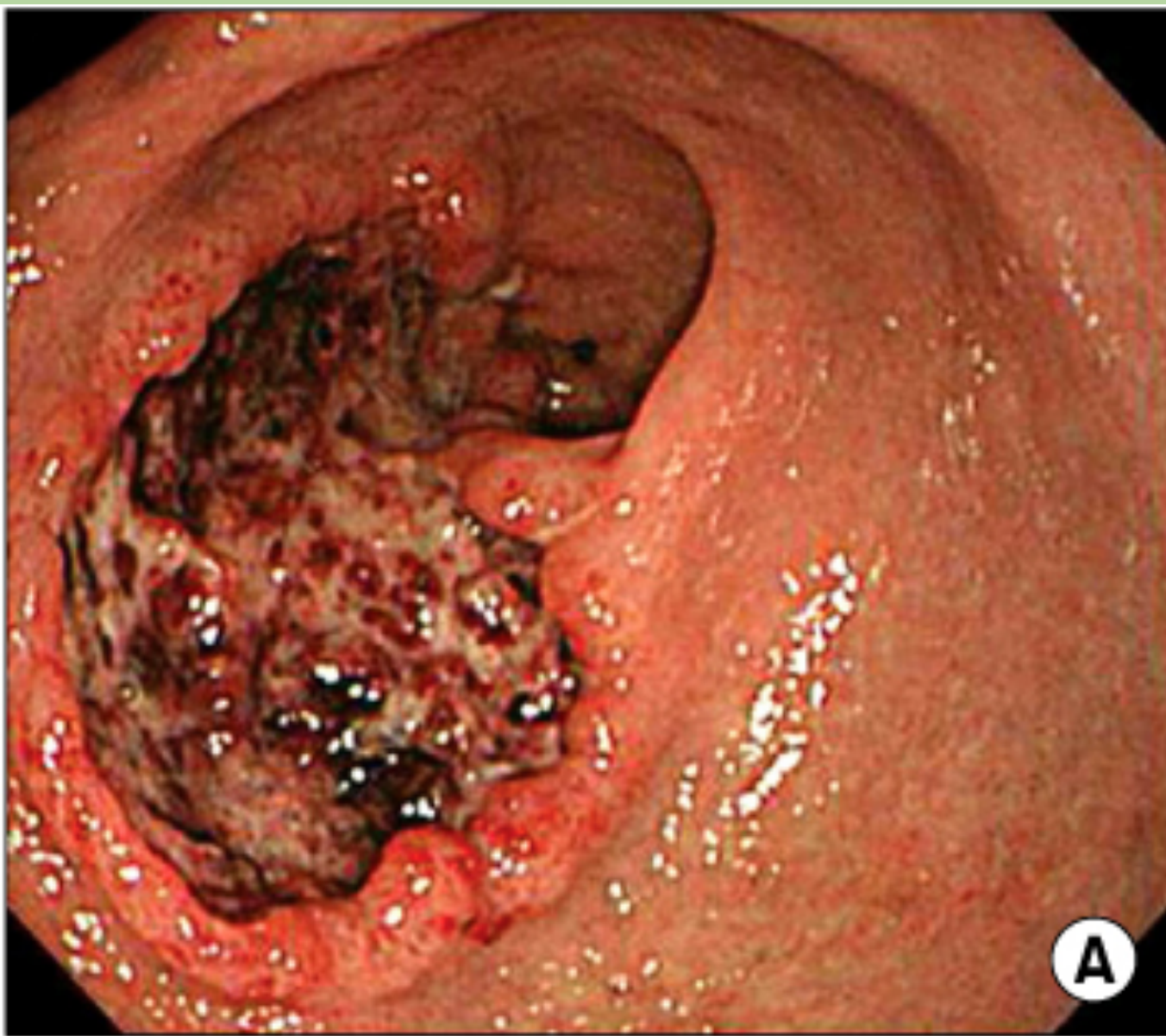
Pyloric antrum of stomach



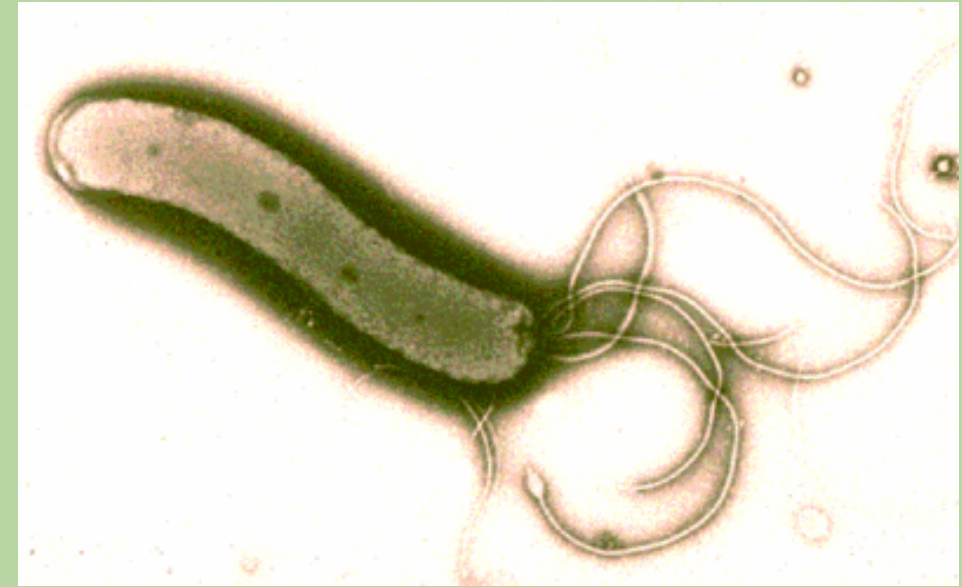
Pyloric sphincter of stomach

Inferior duodenum

Gastric ulcer



Helicobacter pylori



Barry Marshall and Robin Warren – Nobel prize in 2005

Helicobacter pylori

The Clinical Outcomes of *Helicobacter pylori* Infections



Asymptomatic
or chronic gastritis



Chronic atrophic gastritis
Intestinal metaplasia



Gastric or
Duodenal ulcer



Gastric cancer
MALT lymphoma

Gastric carcinoma:

Major risk factors: high salt intake, *Helicobacter pylori* infection (more than 50% of the world's population harbor *H. pylori* in their stomach or duodenum).

Protectives: fresh vegetables and fruits.

Small intestine

- approx. length: 6 m
- **Parts:**
duodenum,
jejunum and ileum

